

Project Summary

1. Project Title

Bike Sharing Demand Analysis

2. Business Problem

Bike-sharing demand varies throughout the day and across different seasons and weather conditions. For management, maintaining the right number of bikes is important because insufficient availability during peak periods can lead to missed rental opportunities, while excess availability during low-demand periods can result in inefficient resource allocation.

The objective of this project is to analyze bike rental patterns and identify **the best season, best time of day, and suitable weather conditions for maximizing bike availability**.

3. Dataset

The project uses the **Seoul Bike Sharing Demand Dataset**, which contains **8,760 hourly records** of bike rental activity along with variables such as hour, date, season, holiday status, temperature, humidity, rainfall, snowfall, and other weather-related information.

The data was analyzed using **Python, Pandas, PostgreSQL, Excel, and Power BI**. Basic web scraping using **BeautifulSoup** was also performed to collect information about Seoul attractions.

4. Data Cleaning

The dataset was prepared using Python and Pandas. The following operations were performed:

- Checked dataset structure, columns, and data types.
- Checked for missing values and duplicate records.
- Checked for invalid data.
- Converted the **Date** column into the appropriate date format.
- Created a **Time_Category** column by grouping hours into Morning, Afternoon, Evening, and Night.
- Created a **Weather Category** column based on rainfall values.
- Exported the cleaned dataset for further analysis using PostgreSQL and Excel.

5. Key Findings

1. Peak Rental Hour

The highest average rental demand occurred at **6 PM**, with **1,502.93 rentals**. The other high-demand hours were 7 PM, 5 PM, 8 PM, and 9 PM, showing that the evening period has the strongest demand.

2. Best Season

Summer had the highest total rental volume, with **2,283,234 rentals**, followed by Autumn, Spring, and Winter.

3. Best Season and Time Combination

The highest-demand combination identified was **Summer + Evening**, with an average of **1,821.33 rentals**. This represents the strongest demand pattern in the analysis.

4. Holiday Effect

Average rentals were higher on **non-holidays (715.23)** compared with **holidays (499.76)**, indicating lower rental activity on holidays within the dataset.

5. Weather Conditions

Demand was highest during **No Rain**, with an average of **739.31 rentals**, compared with **163.84 during Light Rain** and **138.38 during Heavy Rain**. Heavy Rain had only **8 observations**, so this result should be interpreted cautiously.

6. Final Business Question

When should the maximum number of bikes be made available?

Based on the analysis:

- **Best Season:** Summer
- **Best Time of Day:** Evening
- **Peak Hour:** 6 PM
- **Suitable Weather:** No Rain
- **Highest-Demand Combination:** Summer + Evening

Therefore, management should prioritize maximum bike availability during **summer evenings, particularly around 6 PM, under no-rain conditions**.

7. Business Recommendations

Recommendation 1: Increase Availability During Evening Peaks

Management should prioritize bike availability between **5 PM and 9 PM**, particularly at **6 PM**, when rental demand reaches its highest average level.

Recommendation 2: Prepare for Summer Demand

Additional bikes and redistribution efforts should be planned for **summer**, especially during evening hours, as this period shows the strongest overall demand.

Recommendation 3: Adjust Availability Based on Weather

Bike deployment should be adjusted according to weather conditions. Since rental demand is substantially higher during **no-rain conditions**, resources can be redistributed during rainy periods to avoid unnecessary allocation.

Team Information

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